WORKED SOLUTION

STEP II 1987, Question 7

Pure • Trapezium Rule and Factorials

STEP II 1987, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q7

The key idea

The trapezium sum for $\ln x$ with unit steps is almost exactly $\ln n!$, because the heights it adds are $\ln 1, \ln 2, \ldots, \ln n$. Since $\ln x$ is concave, every chord lies below the curve, so the trapezium rule always *underestimates* the integral; that one fact gives the direction of every inequality. For the last part, the trick is to rescale the fine-grid error so that it becomes a unit-step error on a different interval, which can then be compared directly with the error behind $\mathrm{g}(kn)$.

Solution

Justifying the rule

Split $[x_0, x_N]$ into the N strips $[x_r, x_{r+1}]$, each of width h. On each strip, replace the curve by the straight line (chord) joining $(x_r, \mathrm{f}(x_r))$ to $(x_{r+1}, \mathrm{f}(x_{r+1}))$. The region under the chord is a trapezium of area $\frac{1}{2}h\{\mathrm{f}(x_r) + \mathrm{f}(x_{r+1})\}$. Adding over $r = 0, 1, \ldots, N-1$, every interior ordinate $\mathrm{f}(x_1), \ldots, \mathrm{f}(x_{N-1})$ appears in two neighbouring trapezia and the end ordinates appear once each, which gives the stated formula. The approximation is good when h is small, as the chord then stays close to the curve.

Unit steps and $n! \approx \mathrm{g}(n)$

Integrating by parts with $u = \ln x$, $\mathrm{d}v = \mathrm{d}x$,

$$I_n = \int_1^n \ln x \,\mathrm{d}x = \Big[x \ln x\Big]_1^n - \int_1^n 1 \,\mathrm{d}x = n \ln n - n + 1.$$

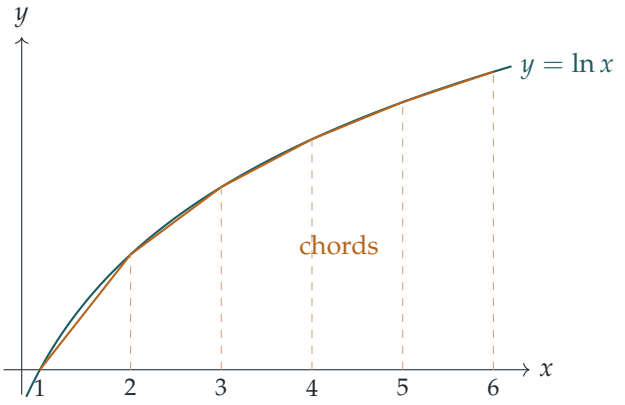
With $h = 1$ the ordinates are at $x = 1, 2, \ldots, n$, so the trapezium sum is

$$\begin{aligned} T_n &= \tfrac{1}{2}\{\ln 1 + 2\ln 2 + \cdots + 2\ln(n-1) + \ln n\} \\ &= \ln 2 + \ln 3 + \cdots + \ln n - \tfrac{1}{2}\ln n = \ln n! - \tfrac{1}{2}\ln n, \end{aligned}$$

using $\ln 1 = 0$. Setting $T_n \approx I_n$,

$$\ln n! \approx (n+\tfrac{1}{2})\ln n + 1 - n, \qquad \text{so} \qquad n! \approx n^{n+\frac{1}{2}}\mathrm{e}^{1-n} = \mathrm{g}(n).$$

Why $n! < \mathrm{g}(n)$



Since $\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}\ln x = -\dfrac{1}{x^2} < 0$, the curve $y = \ln x$ is strictly concave: on each strip $[m, m+1]$ the chord lies strictly below the curve except at its endpoints (see the sketch). So each trapezium has strictly less area than the region under the curve on that strip, and adding the $n-1 \geq 1$ strips,

$$T_n < I_n \quad \Longrightarrow \quad \ln n! - \tfrac{1}{2}\ln n < n\ln n - n + 1 \quad \Longrightarrow \quad \ln n! < \ln \mathrm{g}(n).$$

As exp is increasing, $n! < \mathrm{g}(n)$.

Strips of width $1/k$

Now $h = 1/k$, $N = k(n-1)$ and $x_r = 1 + \frac{r}{k} = \frac{k+r}{k}$ for $r = 0, 1, \ldots, k(n-1)$. The last ordinate is $x_N = n$. Writing the trapezium sum as "h times the sum of all ordinates, minus half of each end ordinate", and using $\ln x_0 = 0$,

$$T_n^{(k)} = \frac{1}{k}\left\{ \sum_{r=0}^{k(n-1)} \ln \frac{k+r}{k} - \tfrac{1}{2}\ln n \right\}.$$

The numerators $k+r$ run through $k, k+1, \ldots, kn$, and there are $k(n-1)+1$ factors of k in the denominators, so

$$\sum_{r=0}^{k(n-1)} \ln \frac{k+r}{k} = \ln \frac{(kn)!}{(k-1)!} - \big(k(n-1)+1\big)\ln k = \ln \frac{(kn)!}{k!} - k(n-1)\ln k,$$

since $(k-1)!\,k = k!$. Setting $kT_n^{(k)} \approx kI_n = kn\ln n - kn + k$ and rearranging,

$$\begin{aligned} \ln(kn)! &\approx \ln k! + k(n-1)\ln k + \tfrac{1}{2}\ln n + kn\ln n + k(1-n) \\ &= \ln k! + \big(kn + \tfrac{1}{2}\big)\ln n + k(1-n)(1-\ln k), \end{aligned}$$

that is,

$$(kn)! \approx k!\, n^{kn+\frac{1}{2}} \left(\frac{\mathrm{e}}{k}\right)^{k(1-n)} =: A,$$

as required.

Which is closer?

Write $E(a,b)$ for the error $\displaystyle\int_a^b \ln t \,\mathrm{d}t$ − (unit-step trapezium sum on $[a,b]$), for integers $a < b$. By the concavity argument above, $E(a,b) > 0$, and errors add over adjacent intervals: $E(1,b) = E(1,a) + E(a,b)$.

The derivation of A was exact apart from replacing $T_n^{(k)}$ by I_n, so

$$\ln A - \ln(kn)! = k\big(I_n - T_n^{(k)}\big).$$

Similarly, from the unit-step work with kn in place of n, $\ln \mathrm{g}(kn) - \ln(kn)! = E(1,kn)$.

Ask yourself How can an error on a grid of width $1/k$ be compared with an error on a grid of width 1?

Substitute $x = t/k$. The ordinates $x_r = (k+r)/k$ become $t = k, k+1, \ldots, kn$, and $\mathrm{d}x = \mathrm{d}t/k$, so

$$k\big(I_n - T_n^{(k)}\big) = \int_k^{kn} \ln\frac{t}{k}\,\mathrm{d}t \;-\; \big(\text{unit-step trapezium sum of } \ln\tfrac{t}{k} \text{ on } [k,kn]\big).$$

Since $\ln\frac{t}{k} = \ln t - \ln k$ and the trapezium rule is exact for the constant $\ln k$, the constant cancels and this is just $E(k,kn)$. Therefore

$$\ln \mathrm{g}(kn) - \ln(kn)! = E(1,kn) = E(1,k) + E(k,kn) = E(1,k) + \big(\ln A - \ln(kn)!\big).$$

Both A and $\mathrm{g}(kn)$ exceed $(kn)!$, and for $k \geq 2$ we have $E(1,k) > 0$, so

$$(kn)! < A < \mathrm{g}(kn) \qquad (k \geq 2).$$

So the new approximation A is closer to $(kn)!$ than $\mathrm{g}(kn)$ is, whether you measure by the difference or the ratio. When $k = 1$ the two are identical, as $A = 1!\, n^{n+\frac{1}{2}}\mathrm{e}^{1-n} = \mathrm{g}(n)$. In words, A uses the *exact* value $k!$ for the first k factors, and only approximates the rest, so it avoids the error $E(1,k)$ from the start of the curve, where ln bends most.

Checks

- $k = 1$ in the final formula gives back $\mathrm{g}(n)$, as it must.
- Take $n = 5$: $5! = 120$ and $\mathrm{g}(5) = 5^{5.5}\mathrm{e}^{-4} \approx 128.0$, so indeed $n! < \mathrm{g}(n)$.
- Take $k = 3$, $n = 5$: $15! \approx 1.3077 \times 10^{12}$, $A \approx 1.3369 \times 10^{12}$ and $\mathrm{g}(15) \approx 1.4102 \times 10^{12}$, confirming $(kn)! < A < \mathrm{g}(kn)$.
- The ratio $\mathrm{g}(kn)/A = \exp E(1,k)$ depends only on k, not n; for $k = 3$ it is $\mathrm{g}(3)/3! \approx 1.0548$, matching the numbers above.

Common slips

- Miscounting the factors of k in the width-$1/k$ sum: there are $k(n-1)+1$ ordinates, and the extra $\ln k$ is what turns $(k-1)!$ into $k!$.
- Arguing "$n! < \mathrm{g}(n)$" from a sketch with the chords above the curve. For a concave function the chords lie below, so the trapezium rule underestimates.
- Forgetting that the error in $\ln A$ is k times the fine-grid error, not the fine-grid error itself.

Beyond STEP

Here $\mathrm{g}(n)$ is a crude form of Stirling's formula, $n! \sim \sqrt{2\pi n}\, n^n \mathrm{e}^{-n}$. Since the errors $E(1,n)$ increase with n and are bounded (each strip's error is at most a constant times $1/m^2$), they converge, which explains why $n!/\mathrm{g}(n)$ tends to the constant $\sqrt{2\pi}/\mathrm{e} \approx 0.922$. The Euler–Maclaurin formula makes this precise and gives the full asymptotic series. Bartle and Sherbert, *Introduction to Real Analysis*, covers the Riemann integral and the error estimates for numerical rules that underpin this.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.